

# Lecture 7

## 1. Machine Translation

- Translating a sentence  $x$  in source language to sentence  $y$  in target language

- **Statistical Machine Translation:**

- so we try to find:

- $$\operatorname{argmax}_y P(y|x)$$

- arg just means returning the sentence(object) of the probability, and given  $x$  output  $y$
  - where we can use **Bayes rule** to break down

- $$\operatorname{argmax}_y P(x|y)P(y)$$

- **Extra: Bayes Model**

- So intuitively, for here, referring to  $x$ , we have **E - Evidence**, which is known, so given  $E$ , we want to find the most likely **hypothesis - H** referring to  $y$  here. which is  $P(y|x)$  or  $P(H|E)$
    - So the Bayes model gives out

$$P(H|E) = \frac{P(H)P(E|H)}{P(E)}$$

- this equation kind of has the intuition of using the answer to find the question, where  $P(E|H)$  is limiting down the evidence with the chosen hypothesis, then multiplying with  $P(H)$  means limiting the chosen hypothesis with its own probability, then dividing over the probability of seeing the evidence, which is  $P(E)$  because it is given, so we choose the ones over it.
  - So back to the MT, the first term  **$P(x|y)$  is the translation model**, models how words and phrases should be translated, given  **$x$  in the target language** and **how likely** is it its translation is  **$x$  in the source language (fidelity)**, and you need to learn from the **parallel data (训练样例)** of the source and target language. And the **second term  $P(y)$  is the language model**, which is the probability of seeing the target generally in english, so represents the **fluency**

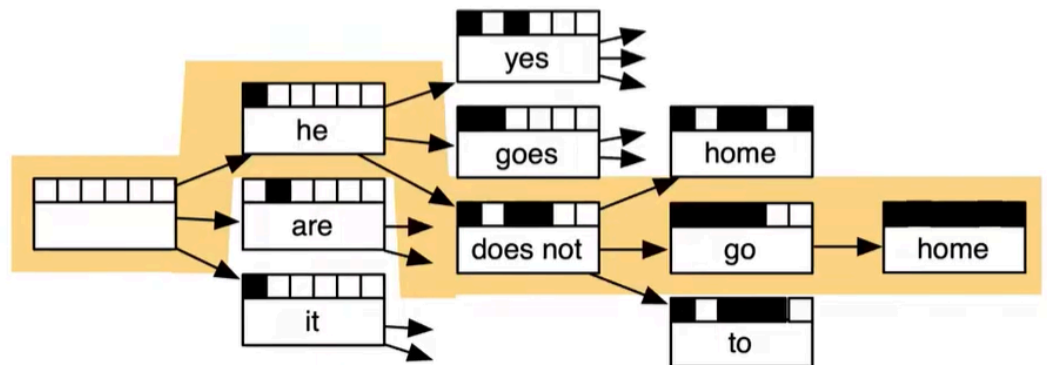
- **Learning of the Translation Model:**

- We can introduce a latent variable  $a$
  - $P(x, a|y)$ ,  $a$  is an **alignment**, where it is the word to word or short phrase translation corresponding between sentence  $x$  and  $y$
  - **Alignment** is complex, some words may not be translated. So it can be many to one or one to many.

- **Decoding:** is the part where the arg function works, with the probability distribution, we find out the target translated sentence

- it works by a dynamic programming process with independence assumption.

| er         | geht         | ja          | nicht     | nach         | hause   |
|------------|--------------|-------------|-----------|--------------|---------|
| he         | is           | yes         | not       | after        | house   |
| it         | are          | is          | do not    | to           | home    |
| , it       | goes         | , of course | does not  | according to | chamber |
| , he       | go           | ,           | is not    | in           | at home |
| it is      |              | not         |           | home         |         |
| he will be |              | is not      |           | under house  |         |
| it goes    |              | does not    |           | return home  |         |
| he goes    |              | do not      |           | do not       |         |
|            | is           |             | to        |              |         |
|            | are          |             | following |              |         |
|            | is after all |             | not after |              |         |
|            | does         |             | not to    |              |         |
|            | not          |             |           |              |         |
|            | is not       |             |           |              |         |
|            | are not      |             |           |              |         |
|            | is not a     |             |           |              |         |



- fill until all the boxes are filled black.

### Destroyed by Neural Machine translation

#### • Neural Machine Translation:

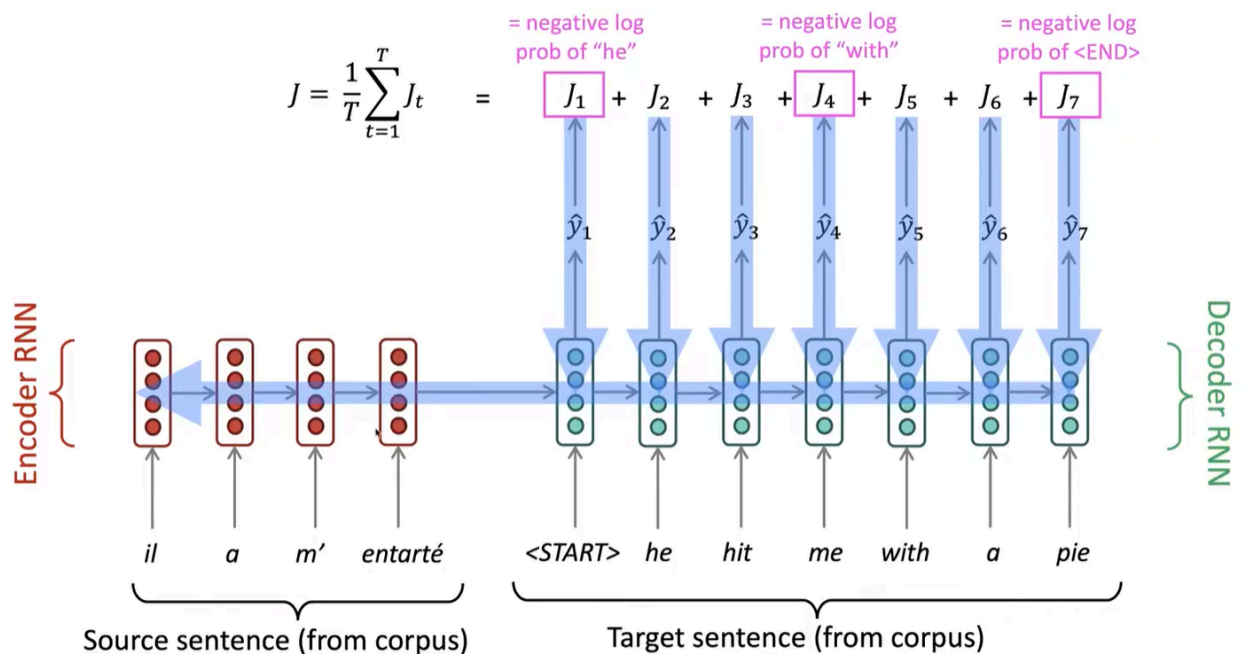
- here, the neural network is referred as **seq-2-seq model**, which involves 2 RNNs, one for the source language to encode, called **Encoder RNN** and one for generating the target language to decode, called **Decoder RNN**. And both of the neural models refers to **Neural Machine Translation (NMT)**
- seq2seq model used in many NLP tasks, such as summarization, parsing...
- seq2seq is a **Conditional language model**, condition refers to the predictions are also conditioned on the input sentence. (which SMT didn't do)

$$P(y | x) = P(y_1 | x) P(y_2 | y_1, x) P(y_3 | y_1, y_2, x) \dots P(y_T | y_1, \dots, y_{T-1}, x)$$

#### ◦ Training NMT

- Still using a large corpus of parallel data.

▪



Seq2seq is optimized as a **single system**. Backpropagation operates “*end-to-end*”.

- generally still same as LSTM RNN training.

## 2. MultiLayer RNN

- by adding multiple RNNs
  - such that more complex representations could be computed
  - lower layer RNNs for low level feature, and vice versa
- also referred as **Stacked RNNs**

## 3. Greedy decoding

- just refers to the process of doing the argmax function where you take the word with the highest probability at each time step
- **Problems:** there is no chance of undoing the decisions made
- **Fix:**
  - **Exhaustive search decoding:** which is tracking all the possible sequences at timeset  $t$ , however too slow and expensive, for a vocab of size  $V$ , it would be complexity of  $O(V^T)$
  - **Beam Search Decoding:** which is tracking the top  $k$  score translations at each time step, which is complexity of  $O(k^T)$ 
    - and the score is calculated by  $\log P_{LM}(y|x)$
    - and higher the score better
    - **Stopping criteria**
      - when reached a prefixed timestep
      - when obtained  $n$  complete hypothesis that is set
    - Then finally out of all the hypothesis, picking the most probable one. But there is a **problem** because the probability formula depends on series of product, so the longer the sequence, the more the product, so **longer sequence may lead to lower scores**.
    - **Fix:** we can normalize the log probabilities by length (negative numbers being divided by larger numbers goes closer to 0/larger)

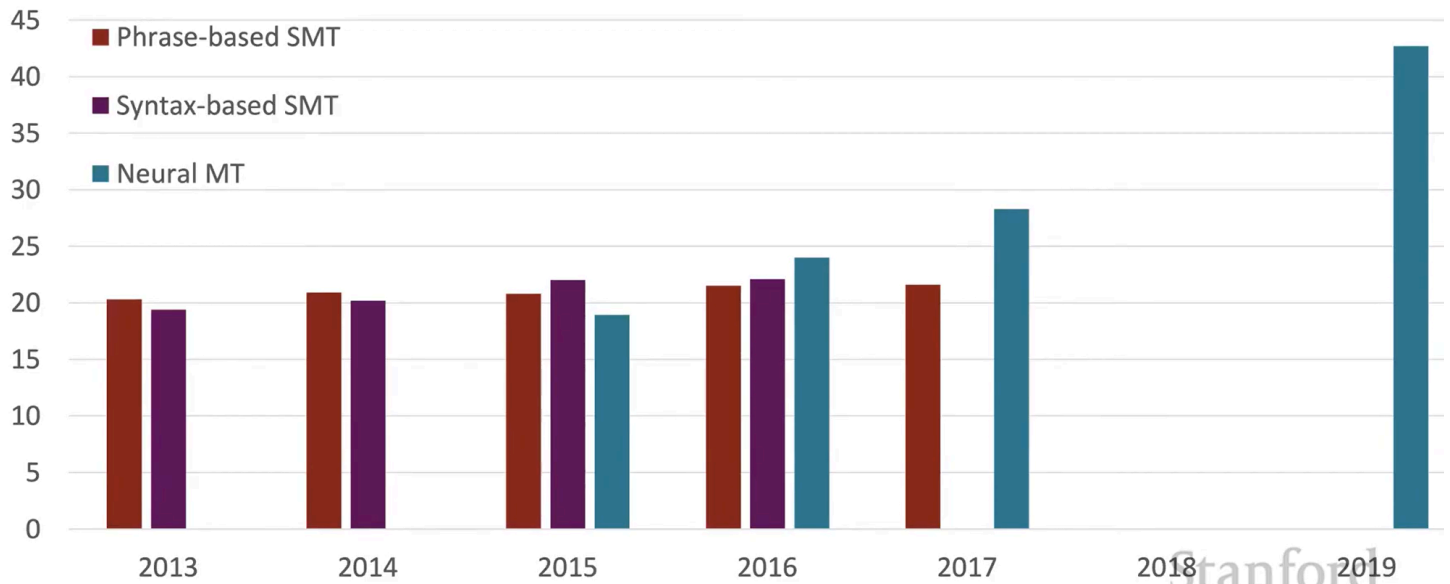
**Overall:** compared to SMT, NMT are more fluent, makes better use of context, better use of phrase similarities.

**however**, they are less interpretable (as using a neural network), and also difficult to control, can't specify rules.

## 4. Evaluation of Machine Translation

- **BLEU score:**

- compares the **machine-written translation** with the **human-written reference translation** and computes similarity score based on
  - n-gram ( $n = 1, 2, 3, 4$ )
  - and a penalty for too short translation. (cuz they give high scores on probability)
  - but still imperfect



- **Eventually** problems still exists

- out of vocab problem
- maintaining context over long text
- pronoun resolution errors (他 - he she him)
- maintaining common sense
- biases also exists.

## 5. Attention Mechanism

- Using a **direct link** between the encoder and decoder, instead of **sequential link** in RNN. So in this way you can focus on a particular part more.

# Sequence-to-sequence with attention

